Predicting Calories Burned During Workout In Gym

1. Project Description:

This project focuses on analyzing gym members' workout and physiological data to understand the factors that influence workout performance and calorie expenditure. The dataset includes various attributes such as age, gender, weight, height, heart rate metrics, workout duration, workout type, body fat percentage, water intake, workout frequency, experience level, and body mass index (BMI).

The project involves several steps including data preprocessing, exploratory data analysis (EDA), feature engineering, and model building. Statistical techniques and visualization methods are used to understand relationships between variables, detect outliers, and identify important features influencing calorie burn.

2. Objective:

The primary objective of the project is to build a machine learning model that can predict the number of calories burned during a workout session based on different physiological and workout-related features.

3. Data collection and Preprocessing:

The dataset used for this project is stored in a CSV file named **gym_members_exercise_tracking.csv** The dataset is read and stored in a variable named **dataset**

```
dataset=pd.read_csv("gym_members_exercise_tracking.csv")
```

```
dataset
```

	Age	Gender	Weight (kg)	Height (m)	Max_BPM	Avg_BPM	Resting_BPM	Session_Duration (hours)	Calories_Burned	Workout_Type	Fat_Percentage	Water_Intake (liters)	Work
0	56	Male	88.3	1.71	180	157	60	1.69	1313.0	Yoga	12.6	3.5	
1	46	Female	74.9	1.53	179	151	66	1.30	883.0	HIIT	33.9	2.1	
2	32	Female	68.1	1.66	167	122	54	1.11	677.0	Cardio	33.4	2.3	
3	25	Male	53.2	1.70	190	164	56	0.59	532.0	Strength	28.8	2.1	
4	38	Male	46.1	1.79	188	158	68	0.64	556.0	Strength	29.2	2.8	
...	...	...	...	...	...	...	...	...	...	...	...	...	
968	24	Male	87.1	1.74	187	158	67	1.57	1364.0	Strength	10.0	3.5	
969	25	Male	66.6	1.61	184	166	56	1.38	1260.0	Strength	25.0	3.0	
970	59	Female	60.4	1.76	194	120	53	1.72	929.0	Cardio	18.8	2.7	
971	32	Male	126.4	1.83	198	146	62	1.10	883.0	HIIT	28.2	2.1	
972	46	Male	88.7	1.63	166	146	66	0.75	542.0	Strength	28.8	3.5	

973 rows × 15 columns

The dataset contains categorical variables such as **Gender** and **Workout_Type**, which cannot be directly used by most machine learning algorithms because they require numerical inputs. To handle these categorical variables, **one-hot encoding** is used

Gender → Gender_Male, Gender_Female

Workout_Type → Workout_Type_Cardio, Workout_Type_Strength, Workout_Type_HIIT, Workout_Type_Yoga

Here **Gender_Female** and **Workout_Type_Yoga** is dropped from the dataset.

```
In [5]: dataset=pd.get_dummies(dataset)
```

```
In [6]: dataset
```

Out[6]:

Duration (hours)	Calories_Burned	Fat_Percentage	Water_Intake (liters)	Workout_Frequency (days/week)	Experience_Level	BMI	Gender_Female	Gender_Male	Workout_Type_Cardio	Work
1.69	1313.0	12.6	3.5	4	3	30.20	0	1	0	
1.30	883.0	33.9	2.1	4	2	32.00	1	0	0	
1.11	677.0	33.4	2.3	4	2	24.71	1	0	1	
0.59	532.0	28.8	2.1	3	1	18.41	0	1	0	
0.64	556.0	29.2	2.8	3	1	14.39	0	1	0	
...	...	...	...	...	...	...	...	...	...	
1.57	1364.0	10.0	3.5	4	3	28.77	0	1	0	
1.38	1260.0	25.0	3.0	2	1	25.69	0	1	0	
1.72	929.0	18.8	2.7	5	3	19.50	1	0	1	
1.10	883.0	28.2	2.1	3	2	37.74	0	1	0	
0.75	542.0	28.8	3.5	2	1	33.38	0	1	0	

```
In [7]: dataset=dataset.drop(columns=['Gender_Female','Workout_Type_Yoga'])
```

Next, dataset is checked for any null value and NA value using the below code,

dataset.isnull().sum()

dataset.isna().sum()

As this dataset has no null value and NA value there is no need to preprocess further.

Now the dataset is ready for univariate and bivariate analysis.

3.1. Univariate Analysis:

Univariate analysis is used to analyze and summarize the characteristics of a single variable in a dataset. It helps in understanding the distribution, central tendency, and variability of the data using statistical measures and visualizations such as histograms or box plots.

3.1.i. Univariate variables:

Here a new dataset named descriptive is created to store all the univariate variables values. It is done by using a for loop as shown below:

```
for columnName in quan:
    descriptive[columnName]["Mean"]=dataset[columnName].mean()
    descriptive[columnName]["Median"]=dataset[columnName].median()
    descriptive[columnName]["Mode"]=dataset[columnName].mode()[0]
    descriptive[columnName]["Q1:25%"]=dataset.describe()[columnName]["25%"]
    descriptive[columnName]["Q2:50%"]=dataset.describe()[columnName]["50%"]
    descriptive[columnName]["Q3:75%"]=dataset.describe()[columnName]["75%"]
    descriptive[columnName]["Q4:100%"]=dataset.describe()[columnName]["max"]
    Q1 = dataset[columnName].quantile(0.25)
    Q3 = dataset[columnName].quantile(0.75)
    IQR = Q3 - Q1
    rule = 1.5 * IQR
    lesser = Q1 - rule
    greater = Q3 + rule
    descriptive.loc["IQR", columnName] = IQR
    descriptive.loc["1.5rule", columnName] = rule
    descriptive.loc["Lesser", columnName] = lesser
    descriptive.loc["Greater", columnName] = greater
    descriptive.loc["Max", columnName] = dataset[columnName].max()
    descriptive.loc["Min", columnName] = dataset[columnName].min()
```

And the dataset,

	Age	Weight (kg)	Height (m)	Max_BPM	Avg_BPM	Resting_BPM	Session_Duration (hours)	Calories_Burned	Fat_Percentage	Water_Intake (liters)	Workout_Frequency (days/week)
Mean	38.6835	73.8547	1.72258	179.884	143.767	62.223	1.25642	905.422	24.9768	2.62662	3.32169
Median	40	70	1.71	180	143	62	1.26	893	26.2	2.6	3
Mode	43	57.7	1.62	198	132	50	1.03	883	28.1	3.5	3
Q1:25%	28	58.1	1.62	170	131	56	1.04	720	21.3	2.2	3
Q2:50%	40	70	1.71	180	143	62	1.26	893	26.2	2.6	3
Q3:75%	49	86	1.8	190	156	68	1.46	1076	29.3	3.1	4
Q4:100%	59	129.9	2	199	169	74	2	1783	35	3.7	5
IQR	21	27.9	0.18	20	25	12	0.42	356	8	0.9	1
1.5rule	31.5	41.85	0.27	30	37.5	18	0.63	534	12	1.35	1.5
Lesser	-3.5	16.25	1.35	140	93.5	38	0.41	186	9.3	0.85	1.5
Greater	80.5	127.85	2.07	220	193.5	86	2.09	1610	41.3	4.45	5.5
Min	18	40	1.5	160	120	50	0.5	303	10	1.5	2
Max	59	129.9	2	199	169	74	2	1783	35	3.7	5

Here are some key inference points from the dataset:

- The average gym member in the dataset is around **38.7 years old**, weighs about **73.85 kg**, and has an average **height of 1.72 m**
- On average, gym members burn around **905 calories per session**, with values ranging from **303 to 1783 calories**, indicating significant variation depending on workout intensity and duration.
- The interquartile range for age is **21 years**, meaning most gym members fall roughly between **28 and 49 years old**
- The most common workout frequency (**mode**) is **3 days per week**, indicating that many members follow a consistent weekly exercise routine
- The maximum heart rate during workouts reaches **199 BPM**, which suggests some members perform **high-intensity workouts.**

3.1.ii. Detecting Outliers:

Outliers are data points in a dataset that **significantly differ from the majority of other observations.** To detect outliers, Q1, Q3 and IQR is used from the above dataset. The formula to find the Upper bound (UB) and Lower bound (LB) is

IQR=Q3-Q1

UB= Q3+1.5*IQR

LB=Q1-1.5*IQR

These values are calculated above in the dataset and stored in Greater and Lesser respectively. Now to detect the columns which have outliers, two lists named lesser and greater is created and the columns are stored as follow:

```
for columnName in quan:
    if(descriptive[columnName]["Min"]<descriptive[columnName]["Lesser"]):
        lesser.append(columnName)
    if(descriptive[columnName]["Max"]>descriptive[columnName]["Greater"]):
        greater.append(columnName)
```

```
lesser
```

```
[]
```

```
greater
```

```
['Weight (kg)', 'Calories_Burned', 'BMI', 'Workout_Type_HIIT']
```

As a result, it is visible that there is no lesser outliers in the dataset. But there are greater outliers in the columns such as Weight (kg), Calories_Burned, BMI and Workout_Type_HIIT.

```
for columnName in greater:
    print(columnName, (dataset[columnName] > descriptive[columnName]["Greater"]).sum())
```

```
Weight (kg) 9
Calories_Burned 10
BMI 25
```

The above image shows that there are 9 outliers in Weight (kg), 10 outliers in Calories_Burned and 25 outliers in BMI. We can omit the outliers present in the column Workout_Type_HIIT as it is a categorical column.

Next is to replace the outliers present in the dataset. It is done by the following lines of code:

```
for columnName in greater:
    dataset.loc[dataset[columnName] > descriptive[columnName]["Greater"], columnName] = descriptive[columnName]["Greater"]
```

3.1.iii. Kurtosis and Skewness:

Kurtosis measures the peaked ness or tail heaviness of a distribution compared to a normal distribution. **Skewness** measures the asymmetry of a data distribution. Both are calculated as follow:

```
for columnName in quan:
    description[columnName]["Kurtosis"]=dataset[columnName].kurtosis()
    description[columnName]["Skewness"]=dataset[columnName].skew()
    description[columnName]["Variance"]=dataset[columnName].var()
    description[columnName]["Standard deviation"]=dataset[columnName].std()
```

```
description
```

	Age	Weight (kg)	Height (m)	Max_BPM	Avg_BPM	Resting_BPM	Session_Duration (hours)	Calories_Burned	Fat_Percentage	Water_Intake (liters)
Kurtosis	-1.21508	-0.0380897	-0.723633	-1.18792	-1.19872	-1.18147	-0.350805	-0.220727	-0.339019	-1.0203
Skewness	-0.077864	0.767964	0.338858	-0.0379505	0.086361	-0.0716359	0.025761	0.224194	-0.635225	0.0714799
Variance	148.375	448.83	0.0163124	132.841	205.782	53.6858	0.117672	73132.9	39.1803	0.360206
Standard deviation	12.1809	21.1856	0.12772	11.5257	14.3451	7.32706	0.343033	270.431	6.25942	0.600172

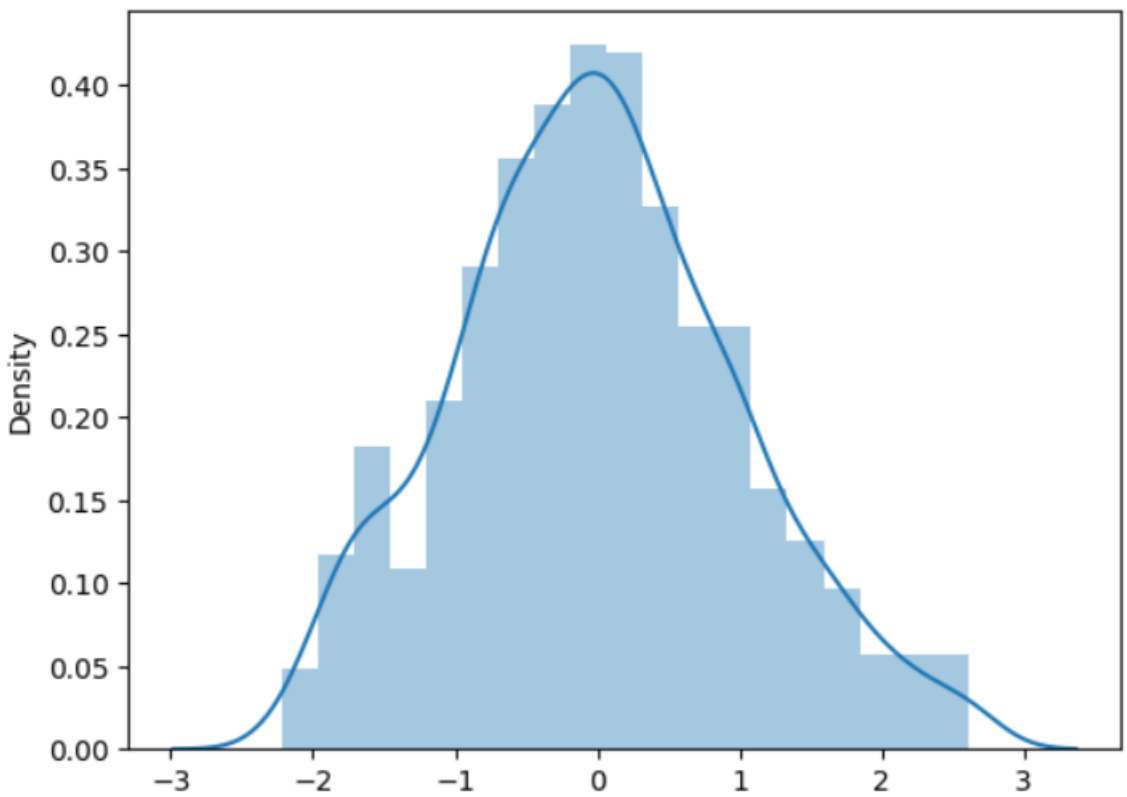
Here are major inference points from the above image attached:

- The skewness value (-1.21) indicates that **Age is approximately normally distributed**, meaning the ages of gym members are fairly balanced.
- Weight shows **positive skewness (0.76)**, suggesting that most members have moderate weight while fewer members have very high weights.

- Session Duration and Water Intake have **positive skewness**, indicating that most members have shorter workout durations and moderate water intake, with a few members having very high values.
- The **high variance (73132) and standard deviation (270.43)** for calories burned show that calorie expenditure varies greatly among gym members.
- **Workout Frequency and Experience Level** show **high kurtosis**, indicating that the data is concentrated around certain values (such as 3–4 workouts per week).

3.1.iv. Standard Deviation:

Standard Deviation is a statistical measure that shows how much the data values deviate or spread from the mean (average). The below graph shows the standard deviation of calories_burned.



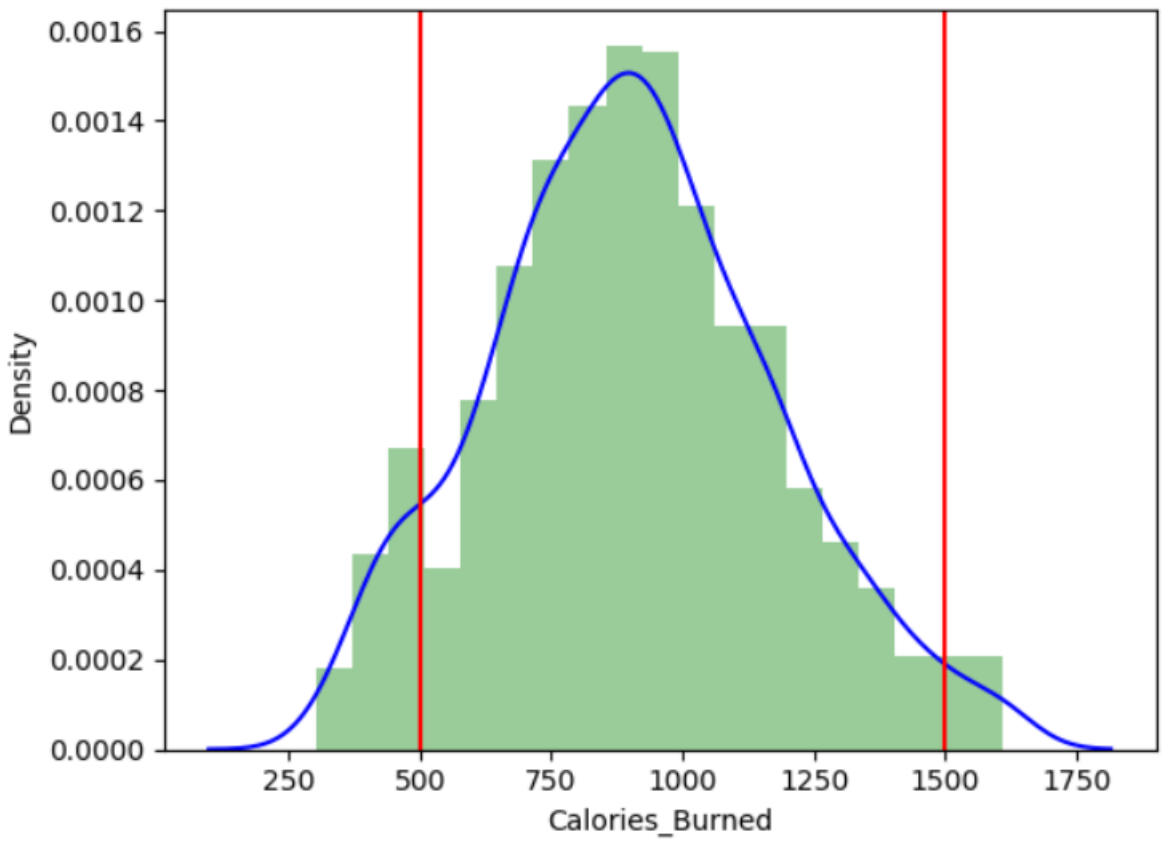
Here are key inference points from the above curve:

- The data follows a **bell-shaped curve**, indicating that the variable is **normally distributed.**
- Most of the data values are concentrated **around the center (near 0)**, showing that the mean and median are close.

- The distribution appears **fairly symmetric**, suggesting **low skewness.**
- Most observations lie between **-1 and 1**, meaning the majority of the data points are close to the average.
- The graph shows a **single peak**, which means the data is **unimodal (one main cluster of values)**

3.1.v. Probability Density Function:

The **Probability Density Function (PDF)** describes the likelihood of a continuous random variable taking values within a particular range. The below graph shows the PDF for calories_burned in the range of 500 to 1500:



3.2. Bivariate Analysis:

Bivariate analysis is the statistical analysis of two variables to determine the relationship or association between them. It helps understand how one variable changes with respect to another variable using methods like correlation, scatter plots, or hypothesis tests.

3.2.i. Correlation and Covariance:

Correlation is the relation between two columns of the dataset and Covariance is the difference between two columns. The correlation and covariance between the columns of the dataset and Calories_burned is calculated as follow:

```
bivariate=pd.DataFrame(index=quan,columns=["Covariance","Correlation"])
```

```
for columnName in quan:
    bivariate["Covariance"][columnName]=dataset["Calories_Burned"].cov(dataset[columnName])
    bivariate["Correlation"][columnName]=dataset["Calories_Burned"].corr(dataset[columnName])
```

And the result is,

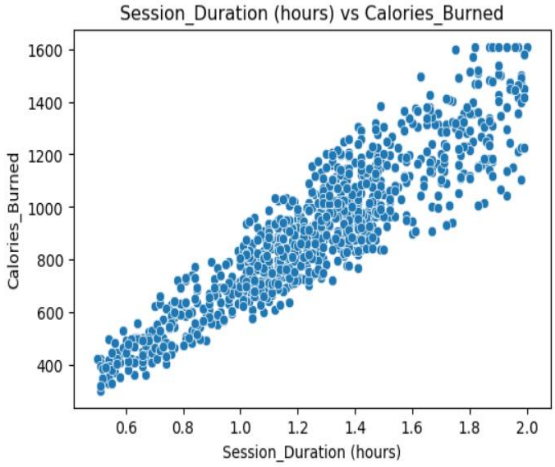
	Covariance	Correlation
Age	-505.715	-0.153522
Weight (kg)	544.312	0.0950059
Height (m)	2.94439	0.0852472
Max_BPM	3.01223	0.000966418
Avg_BPM	1312.79	0.338403
Resting_BPM	29.4639	0.0148698
Session_Duration (hours)	84.406	0.909872
Calories_Burned	73132.9	1
Fat_Percentage	-1008.98	-0.596062
Water_Intake (liters)	57.7302	0.355689
Workout_Frequency (days/week)	142.413	0.576769
Experience_Level	139.049	0.695121
BMI	108.206	0.0624084
Gender_Male	20.1625	0.149226
Workout_Type_Cardio	-5.29447	-0.0444963
Workout_Type_HIIT	4.52305	0.0398988
Workout_Type_Strength	1.12743	0.00943978

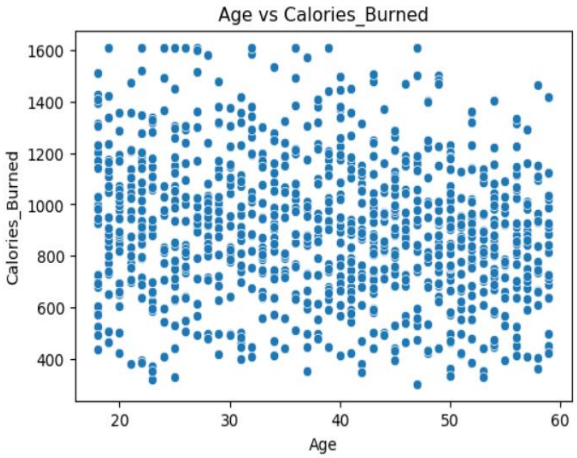
Here are key inference points from the above image:

- **Session_Duration (hours)** has the **strongest positive correlation (0.909)** with Calories_Burned, meaning longer workout sessions burn more calories.
- **Experience_Level (0.695)** and **Workout_Frequency (0.577)** also show **strong positive relationships**, indicating that more experienced and frequent exercisers burn more calories.
- **Age (-0.153)** has a **weak negative relationship**, meaning older individuals may burn slightly fewer calories.

3.2.ii. Scatter plots:

A scatter plot is a graphical representation used to show the relationship between two numerical variables. As Session_Duration and Age has a strong relationship with calories_burned, the scatter plot is plotted between Session_Duration, Age and calories_burned respectively.





3.2.iii. Detecting Multi-collinearity:

Multi-collinearity occurs when independent variables are highly correlated with each other. To detect multicollinearity, Variance Inflation Factor (VIF) is used.

Before calculating VIF for the dataset we should delete the dependent variable (calories_burned) from the dataset. As the target variable is expected to correlate with features, including the target in VIF gives meaningless results.

Also we are dropping Gender_Male, Workout_Type_Cardio, Workout_Type_HIIT and Workout_Type_Strength as these are categorical values.

	variables	VIF
0	const	4483.904762
1	Age	1.014917
2	Weight (kg)	71.656916
3	Height (m)	20.565025
4	Max_BPM	1.012520
5	Avg_BPM	1.008776
6	Resting_BPM	1.009894
7	Session_Duration (hours)	2.506726
8	Fat_Percentage	2.588810
9	Water_Intake (liters)	1.867429
10	Workout_Frequency (days/week)	3.365744
11	Experience_Level	5.311713
12	BMI	64.272242

From the above image it is visible that height, weight and BMI has high VIF. This is because BMI is usually calculated by using the below formula.

$$BMI = \frac{Weight\ (kg)}{Height\ (m)^2}$$

Hence we are dropping the BMI column from the dataset.

3.2.iv. T-Test:

T-Test is used to compare a numeric variable between two groups. The result of T-Test will provide p-value. This p-value is used to find the similarity between groups based on mean.

If p-value < 0.05 -> Reject Null Hypothesis (Accept Alternate Hypothesis)

If p-value > 0.05 -> Accept Null Hypothesis (Reject Alternate Hypothesis)

Null Hypothesis – There is no significant relationship between groups

Alternate Hypothesis - There is a significant relationship between groups

3.2.iv.a. Unpaired T-Test:

Different Group (Male, Female) but same condition (Calories_burned)

```
from scipy.stats import ttest_ind
male=dataset[dataset['Gender_Male']==1]['Calories_Burned']
female=dataset[dataset['Gender_Male']==0]['Calories_Burned']
ttest_ind(male,female)
```

```
Ttest_indResult(statistic=4.70267792129786, pvalue=2.9391772789462282e-06)
```

Here p-value is less than 0.05. We can reject null Hypothesis. Hence, mean Calories_Burned is the different for males and females.

3.2.iv.b. Paired T-Test:

Same Group (Male) but different condition (Calories_Burned, Session_Duration (hours))

```
from scipy.stats import ttest_rel
calories=dataset[dataset['Gender_Male']==1]['Calories_Burned']
session=dataset[dataset['Gender_Male']==1]['Session_Duration (hours)']
ttest_rel(calories,session)
```

```
Ttest_relResult(statistic=75.34938071527083, pvalue=1.445568117380763e-278)
```

Here p-value is less than 0.05. We can reject null Hypothesis. Hence, if males spend more time in gym calories_burned is more.

3.2.v. ANOVA test:

ANOVA (Analysis of Variance) is a statistical test used to determine whether there are significant differences between the means of three or more groups.

```
cardio = dataset[dataset["Workout_Type_Cardio"] == 1]["Calories_Burned"]
hiit = dataset[dataset["Workout_Type_HIIT"] == 1]["Calories_Burned"]
strength = dataset[dataset["Workout_Type_Strength"] == 1]["Calories_Burned"]
from scipy.stats import f_oneway
f_stat, p_value = f_oneway(cardio, hiit, strength)
p_value
```

```
0.258082975720694
```

Here p-value is greater than 0.05. We can accept null Hypothesis. There is no significant difference. Hence, Mean Calories_Burned is equal across all workout types.

4. Feature Selection

Feature selection is used to choose the most important input variables (features) that help predict the output variable. It improves model accuracy, reduces overfitting, and decreases computation time.

Here, **SelectKBest Method** is used

```
from sklearn.feature_selection import SelectKBest
from sklearn.feature_selection import f_regression
input_feature = dataset.drop(columns=["Calories_Burned"])
output_feature = dataset["Calories_Burned"]
selector = SelectKBest(score_func=f_regression, k=5)
X_new = selector.fit_transform(input_feature, output_feature)
selected_features = input_feature.columns[selector.get_support()]
print(selected_features)
```

```
Index(['Session_Duration (hours)', 'Fat_Percentage', 'Water_Intake (liters)',
       'Workout_Frequency (days/week)', 'Experience_Level'],
      dtype='object')
```

Hence, For this **Calories_Burned prediction** dataset, the most useful features are:

- Session_Duration (hours)
- Fat_Percentage
- Water_Intake (litres)
- Workout_Frequency (days/week)
- Experience_Level

5. Model Creation:

The first step of creating the model is problem identification. Here,

Stage-1: Domain Selection – Machine Learning

Stage-2: Learning Selection – Supervised Learning

Stage-3: Supervised – Regression

There are various Regression algorithms available. R-score is used for evaluation of the model. We have created model using various algorithms and the results are given below,

Algorithm	R-score
Linear Regression	0.8647
Decision Tree	0.2122
Random Forest	0.8155

The regression model achieved an R^2 score of **0.8647**, indicating that approximately **86% of the variance in calories burned is explained by the selected independent variables**, demonstrating a strong predictive performance. Hence the linear regression model is created and the same is saved in a file named as Predicting_Calories_Burned_During_Workout.sav

6. Deployment:

A new python notebook is created and the model is loaded in a variable using pickle. The input features values are entered by the user and stored in variables. These variables are given as input for the loaded model and the result is printed as follow,

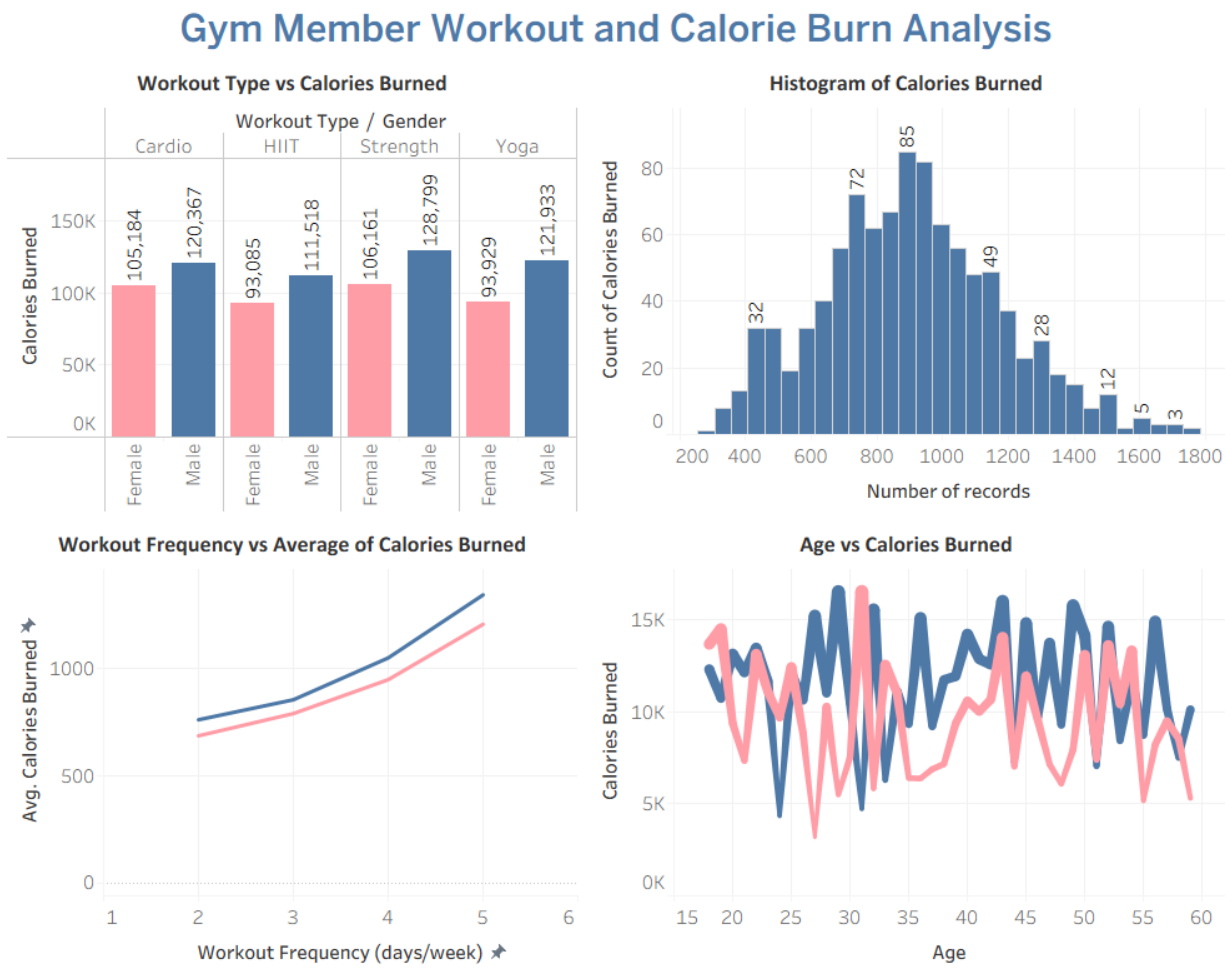
```
new_data = pd.DataFrame([[session,fat,water,day,exp]],
                        columns=['Session_Duration (hours)',
                                 'Fat_Percentage',
                                 'Water_Intake (liters)',
                                 'Workout_Frequency (days/week)',
                                 'Experience_Level'])
prediction = loaded_model.predict(new_data)
```

```
print("The amount of Calories burned is:",prediction[0])
```

```
The amount of Calories burned is: 1422.2877120989965
```

7. Tableau:

Tableau is used to visualize relationships between workout duration, frequency, age, and calories burned. Here a dashboard is created which is displayed below,



The dashboard consists of four different graphs which are explained below:

Graph 1: Workout Type vs Calories Burned

This bar chart compares the total calories burned for different workout types and gender.

- Male members burn **more calories than female members** across all workout types.
- **Strength training** shows the **highest calories burned**, especially for males (~128K).
- Cardio, HIIT, and Yoga show slightly lower but comparable calorie burn.

Workout intensity and gender influence the total calories burned, with males generally recording higher calorie expenditure.

Graph 2: Histogram of Calories Burned

This histogram shows the distribution of calories burned among gym members.

- Most members burn **between 700 and 1100 calories per session**.
- The peak of the distribution occurs around **850–950 calories**.
- Very few members burn extremely low (<400) or extremely high (>1500) calories.

Calories burned follow a roughly normal distribution, meaning most workouts produce moderate calorie expenditure.

Graph 3: Workout Frequency vs Average Calories Burned

This line chart shows the **relationship between weekly workout frequency and average calories burned**.

- As workout frequency increases from **2 days to 5 days per week**, the **average calories burned also increases**.
- Both male and female members show this upward trend.
- Males consistently burn slightly more calories than females.

Higher workout frequency is associated with **greater calorie expenditure**, indicating that consistent exercise leads to higher energy burn.

Graph 4: Age vs Calories Burned

This line graph shows the variation of calories burned across different ages.

- Calories burned fluctuate across ages from **18 to 60**.
- Some peaks appear in the **late 20s to early 40s**.

This graph shows almost linear relationship with no major increase or decrease in the calorie expenditure based on age.

8. Project Findings and Results:

The Gym Members Exercise Tracking dataset was analyzed to study the factors influencing calories burned during workouts. The dataset was first loaded and preprocessed by converting categorical variables such as **Gender and Workout Type** into dummy variables to make the data suitable for analysis.

Univariate analysis was performed using descriptive statistics such as mean, median, quartiles, and standard deviation to understand the distribution of the variables. Outliers were detected using the **IQR method** and treated accordingly. Skewness and kurtosis were also calculated to examine the shape of the data distribution.

Bivariate analysis was conducted using covariance, correlation, and visualizations such as scatter plots and histograms to identify relationships between variables. The analysis showed that **Session Duration, Workout Frequency, and Experience Level** have a strong influence on **Calories Burned**.

Feature selection method **SelectKBest** were used to identify the most relevant predictors. Multicollinearity was checked using **Variance Inflation Factor (VIF)** to ensure the reliability of the regression model.

Finally, machine learning models were developed to predict **Calories Burned**. Among the tested models, **Linear Regression achieved an R^2 score of approximately 0.86**, indicating strong predictive performance. Interactive dashboards were also created using **Tableau Desktop** to visualize key insights and patterns in the data.

The results show that workout duration, frequency, and experience level significantly affect the number of calories burned during exercise.